Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–February 16th, 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Are PRAM architectures using explicit or implicit synchronizations? (1)

B. Please define what is meant by "Speed up" in a PRAM application. (1)

C. Please describe how prefix scan works when a PRAM architecture is considered. (1)

D. Is the Amdahl law a fixed-time model? (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define when Las Vegas and Monte Carlo algorithms are defined as efficient algorithms. (2)

B. What are the key properties that the auxiliary sorting algorithm used by the radix sort must have to achieve efficient and correct sorting? (1) **This question can be skipped in case of a passed challenge.**

C. Please describe the divide and conquer powering of a number algorithm. What complexity does it have? (1)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss how the longest common subsequence works. What about its complexity? (2)

B. Please provide the implementation of the union operation for Treaps. What is the operation complexity? (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. What are the key differences between Karger vs Karger&Steing algorithms? (2)

B. What is meant by "accounting method"? To which class of algorithms could be applied? Examples? (2)